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Sub ject : DSA

Unit 3, S7, SLO2 - Implementation of Stack using Linked List

Build the code for Implementation of Stacks using Linked Lists (Pointer Implementation):

a. Define the structure for a node

b. Write a function for push operation with arguments top and int val

c. Write a function for pop operation with arguments top

d. Write a function for peek operation with arguments top

Answer: /* a. Node Structure Definition */

```
struct node {  
    int data;  
    struct node *next;  
};
```

/* b. Push Function */

```
struct node *push(struct node *top, int val) {  
    struct node *new_node = (struct node *)malloc(sizeof(struct node));  
    if (new_node == NULL) {  
        printf("  
Heap Overflow! Unable to allocate memory\n");  
        return top;  
    }  
    new_node->data = val;  
    new_node->next = top;  
    top = new_node;  
    printf("  
Pushed %d onto linked stack\n", val);  
    return top;  
}
```

/* c. Pop Function */

```
struct node *pop(struct node *top) {  
    if (top == NULL) {  
        printf("  
Stack Underflow! Linked stack is empty\n");  
        return NULL;  
    }  
    struct node *ptr = top;  
    printf("  
Popped element: %d\n", ptr->data);  
    top = top->next;  
    free(ptr);  
    return top;  
}
```

/* d. Peek Function */

```
int peek(struct node *top) {  
    if (top == NULL) {  
        printf("  
Stack is Empty!\n");  
        return -1;  
    }  
    return top->data;  
}
```